Zhan Pengyu

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EDUCATION

Peking University Health Science Center (PKUHSC) — Beijing, China
Candidate for M.D. in Clinical Medicine — Sept 2022 – July 2027 (Expected)

• **Key Academic Highlights (Core Competencies):**

- **Computer Science & Engineering:** Introduction to Computing (94/100), General Physics (92/100), Electronic Info Tech Practice (91/100), Rapid Prototyping (89/100).
- **Research & Clinical Reasoning (PBL):** Innovation Comprehensive Experiment (97/100), Innovation Thinking III (93/100), General Surgery (88/100).

• English Proficiency: TOEFL iBT: 4.5/6.0 (completed June 2026)

RESEARCH INTERESTS

- Multimodal Learning in Healthcare & Medical Foundation Models (VLM/LLM);
- Clinical Reasoning Logic powered by Artificial Intelligence;
- Medical Vision & Generative AI (2D-to-3D Reconstruction, Point Cloud Processing);
- Surgical Robotics & Precision Navigation (Translational Research & Tech-transfer).

RESEARCH EXPERIENCE

Orthopedic Robotics & AI Research Group — *Aerospace Center Hospital*
Core Student Investigator & Project Lead — *April 2025 – Present*

Project 1: AI-Driven 3D-Printed Surgical Guide System (Joint w/ BIT)

Overview: Spearheading the development of personalized 3D-printed surgical guides for UKA/ACL procedures, acting as a bridge between clinical pain points and multimodal algorithmic solutions.
Role & Execution: Serving as Clinical Tech Coordinator. Leading a cross-functional team to integrate AI-based bone segmentation with rapid prototyping (3D printing) to achieve robotic-level precision at a lower cost.
Cross-Disciplinary Collaboration: Collaborating directly with the Multi-Modal Lab at Beijing Institute of Technology (BIT) for algorithm validation and technical implementation. Competed in 2026 Innovation Beijing Challenge Cup.

Project 2: Robotic-Assisted UKA (Translational Research)

Overview: Leading a translational study evaluating the surgical accuracy and learning curve of the Changmugu™ Robotic System (Fixed-Bearing Platform).
Primary Operator: Served as the primary student operator in a supervised research setting, completing 20 robotic UKA procedures on sawbone models. Mastered the full workflow: CT-based planning, registration, and robotic arm control.
Study Design & Analysis: Designed the experimental protocol to quantify surgical precision (bone resection accuracy & mechanical alignment). Analyzed data from 40 cases to establish the learning curve.
Outcome: Manuscript drafted and finalized. Submitted to *Frontiers in Bioengineering and Biotechnology* (Biomechanics). **First author.**

Project 3: Robotic-Assisted ACL Reconstruction (2026 NSFC Grant Preparation)

Study Type: A prospective multi-center RCT in collaboration with Beijing Jishuitan Hospital (National Center for Orthopedics).
Role: Student Co-Investigator (Assisting PI: Dr. Tang Zhengjie).
Contribution: Formulated the Randomized Controlled Trial (RCT) protocol (n=120) and defined the technical roadmap for validating TiRobot-assisted tunnel accuracy.
Strategic Work: Currently synthesizing preliminary data (from sawbones/cadavers) to demonstrate feasibility and support the 2026 National Natural Science Foundation of China (NSFC) grant application.

Plastic Surgery AI Lab — *Peking University Third Hospital*
Research Assistant & Algorithm Developer — *Sept 2024 – May 2026*

Project: AI-Driven 3D Reconstruction & Clinical Research (Prof. An Yang)

First Formal Research Group: Weekly journal clubs covering the full spectrum of plastic surgery research — basic science, clinical study design, surgical techniques (rhinoplasty), and craniofacial imaging. Built foundational knowledge through intensive literature immersion.
Key Publication (JPRAS 2025): Co-authored AI-guided navigation paper for CT-based safe harvesting area identification.

Independently annotated 1/3 of patient nasal septum CT scans using Mimics & 3D Slicer — first exposure to medical image segmentation, STL file processing, and machine learning workflows.
Additional Contributions: Assisted in editing a plastic surgery textbook, narrated surgical technique videos, and co-authored numerous peer reviews — developing academic writing and critical appraisal skills.
Independent Exploration: Built a proof-of-concept Diffusion Model pipeline (PyTorch) for 2D-to-3D point cloud generation from medical images, bridging clinical imaging with generative AI.

Independent & Collaborative Research Projects

Multi-Institutional

Research Lead & Collaborator — *2026.3 – Present*

Project: Breast Tumor Localization — Prone MRI → Supine Prediction
Overview: Developing a novel clinical pipeline combining ICP rigid alignment, CPD non-rigid deformation, and PointNet++ deep learning to predict supine breast tumor position from prone MRI. **First author.**
Data & Scale: Curated and processed 314 paired cases from Shenzhen No.2 Hospital — among the largest datasets in this domain. Targeting JBHI 2026.
Methods: Surface point cloud registration → non-rigid deformation → deep learning prediction. Full pipeline implementation in PyTorch.

Project: LLM Long-term Memory via Dynamic Knowledge Graphs
Overview: Building on temporal knowledge-graph frameworks, researching improvements to episodic memory for LLM agents: entity deduplication, temporal decay strategies, and multi-hop RAG retrieval over evolving graphs.
Target: ICLR 2027 submission. Architecture design and prototype in progress.

Project: Embodied AI — Motion Recognition & Assessment (w/ BIT Multi-Modal Lab, Prof. Ma Kang)
Overview: Developing multi-modal sensor fusion pipelines for motion recognition and scoring, with applications in surgical skill assessment and rehabilitation monitoring.

Project: ADHD & School Refusal — Multi-Family Group Therapy RCT (PKU Sixth Hospital, Prof. Qian Ying)
Role: Clinical Research Intern (July – Oct 2024). Immersed in adolescent psychiatry outpatient clinic, learning clinical research methodology from one of China's leading psychiatric institutions.
Core Learning: Mastered statistical analysis methods (SPSS), meta-analysis workflows, and clinical trial design through weekly journal clubs. Psychiatry's rigorous statistical tradition shaped my understanding of evidence quality and methodology–results interpretation.
Contribution: Participated in RCT data collection and organization, presented latest MFGT and school refusal intervention literature. Co-authored the resulting *BMC Psychology* (2026) publication.

PUBLICATIONS

Core Medical AI & Robotics:

[1] **Zhan P**, et al. Robotic-Assisted UKA Surgical Accuracy and Learning Curve Analysis. *Submitted to Frontiers in Bioengineering and Biotechnology* (Biomechanics). **(First author)**

[2] Shen H*, Pan X*, Du S*, **Zhan P**, et al. AI-based three-dimensional identification of safe harvesting area of perpendicular plate of ethmoid in rhinoplasty. *Journal of Plastic, Reconstructive & Aesthetic Surgery (JPRAS)*, 2025.

Collaborative Clinical Research:

[3] Wang Y, Wang X, Xiong Y, **Zhan P**, Ding R, Shi C. Effectiveness of Multi-Family Group Therapy for School Refusal in Adolescents with Depressive Disorders: Study Protocol for a Randomized Controlled Trial. *BMC Psychology*, 2026.

[4] **Zhan P**, et al. Atrial septal pacing reduces atrial fibrillation compared with right atrial appendage pacing: Insights from the clinical evidence. *Manuscript in preparation*.

[5] **Zhan P** (Editor). *Understanding Pneumothorax: Mastering the Secrets of Recovery*. Medical Popular Science Book, published.

SKILLS

Programming & AI/ML

- **Python, C++** — proficient since middle school; primary languages for research
- **Deep Learning:** PyTorch, PointNet++, Diffusion Models, ICP/CPD, 3D Point Cloud Processing
- **LLM & AI Tools:** Claude Code (advanced), OpenAI Codex (advanced), Git, GraphRAG, Knowledge Graphs
- **Medical Image Analysis:** CT/MRI/DICOM, 3D Reconstruction, Computer Vision

Clinical & Translational Research

- **Surgical Robotics:** Changmugu, TiRobot — operation, CT planning, CUSUM learning curve analysis
- **Clinical Research:** RCT protocol design (n=120), systematic review, clinical data management
- **Translational:** Clinical needs analysis, PoC design, interdisciplinary team coordination, NSFC grant preparation
- **Rapid Prototyping:** 3D printing, surgical guide design, medical device innovation